

- 1 Epithelial cells in the small intestine have cell structures known as microvilli. The microvilli of these cells are found only on the surface that borders the gut lumen.

Fig. 1.1 shows images of microvilli of intestinal epithelial cells. These images have been obtained using a scanning electron microscope and a transmission electron microscope.

Fig. 1.1A is at a different magnification to Fig. 1.1B.

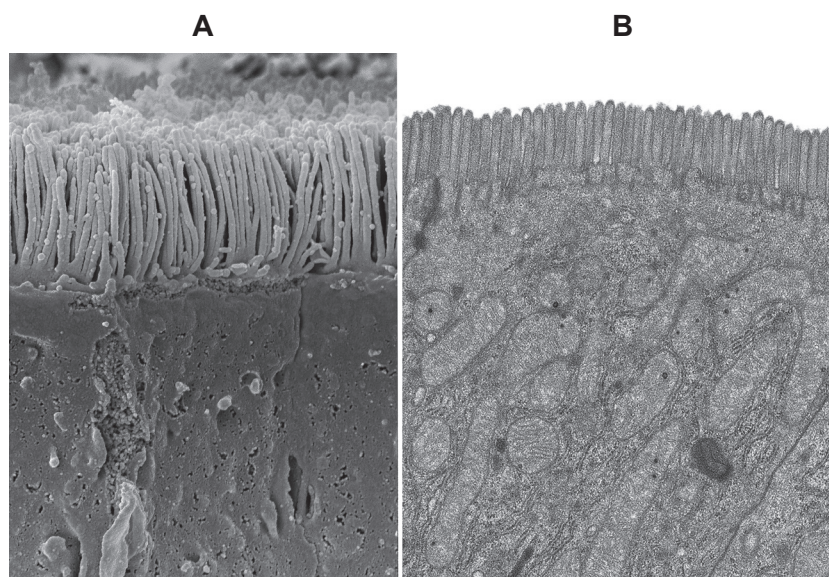


Fig. 1.1

- (a) With reference to Fig. 1.1, state how it is possible to distinguish between a scanning electron micrograph and a transmission electron micrograph.

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- (b) The approximate length of a microvillus is $1\text{ }\mu\text{m}$.

Outline the method you would use to estimate the magnifications of the images shown in Fig. 1.1.

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- The events occurring in an intestinal epithelial cell during the absorption of glucose are summarised in Fig. 1.2.



- Sodium ions (Na^+) are removed from the cell by active transport through a transport protein known as a sodium-potassium (Na^+/K^+) pump.
 - This decreases the concentration of Na^+ in the cell compared to the gut lumen.
 - Glucose molecules are cotransported with Na^+ into the cell from the gut lumen.
 - Glucose molecules are transported out of the cell into the tissue fluid down a concentration gradient.
- (i) Active transport involves water-soluble substances, such as Na^+ and K^+ , and the use of ATP to provide the energy needed for their transport through carrier proteins.

Outline **other** features of active transport.

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- (ii) Glucose molecules enter the cell through a membrane protein.

Suggest why glucose molecules need to be cotransported with Na^+ when it enters the cell through the membrane protein.

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- (iii) Explain how microvilli increase the uptake of glucose into an intestinal epithelial cell.

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- (d) Stem cells are also located in the wall of the small intestine. These cells divide by mitosis continuously.

Suggest **and** explain the importance of mitosis by stem cells in the small intestine.

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